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1. Unsupervised Learning

1.1 Problem Statement

We aim to develop a model that can identify relationships within the Wine Quality dataset using physicochemical properties such as acidity, alcohol content, sugar levels, in order to uncover whether wines can be meaningfully grouped by their physicochemical profiles and assess if these groupings provide more insights than predefined quality labels. Objective chemical characteristics have been shown to significantly influence perceived wine quality, with attributes such as alcohol, sulphates, and fixed acidity identified as major determinants in distinguishing quality levels among wines (Amalia, Valerio, Cinzia, Carmine & Luigi, 2023).

1.2 Research Questions (RQ)

- **RQ1:** Which is the most suitable number of clusters when grouping wines according to their physicochemical properties?
- **RQ2:** What insights can be derived from the clusters?
- **RQ3:** How do the identified clusters relate to observed wine quality levels, and do they reveal meaningful patterns not captured by individual variables alone?

1.3 Dataset Description and Exploratory Data Analysis (EDA)

Key Features Explanations

Fixed Acidity: Concentration of non-volatile acids	Volatile Acidity: Concentration of volatile acids	Citric Acid: Concentration of Citric Acid	Residual sugar: Amount of sugar remaining in wine after fermentation is complete.
Chlorides: Amount of sodium chloride in the wine.	Free sulfur dioxide: Amount of sulfur dioxide that is not bounded to other compounds.	Total sulfur dioxide: Total amount of sulfur dioxide, including both free and bound forms.	Density: Mass per unit volume of wine
pH: Acidity level of the wine	Sulphates: Amount of sulphates compounds added.	Alcohol: Alcohol content of the wine.	Quality: Target variable representing wine quality.

The Wine Quality dataset comprises of 1018 rows and 12 columns after data cleaning*. The correlation matrix (Fig 1.0)* shows a strong positive association between alcohol and quality, indicating that wines with greater alcohol content typically attain better quality ratings. Conversely, volatile acidity exhibits a moderate negative correlation with quality, suggesting that increased volatile acidity is linked to lower perceived wine quality. These relationships provide a basis for feature selection in clustering, as variables closely related to quality may contribute to the formation of distinct and meaningful clusters that reflect underlying physicochemical differences among wines.

The scatterplot, Relationship between Alcohol and Quality (Fig1.1)*, demonstrates a clear positive linear relationship between alcohol and quality, with greater alcohol levels typically linked to better quality scores, despite some variability. The horizontal clustering of points reflects the discrete nature of the quality variable, which is recorded as integer ratings rather than continuous values. Collectively, the final outcome suggests that alcohol is a key predictor of wine quality.

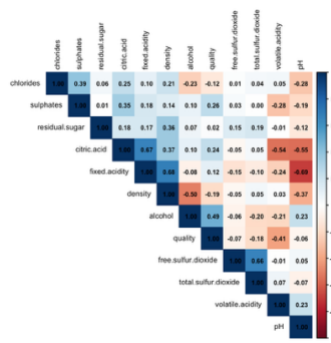


Figure 1.0

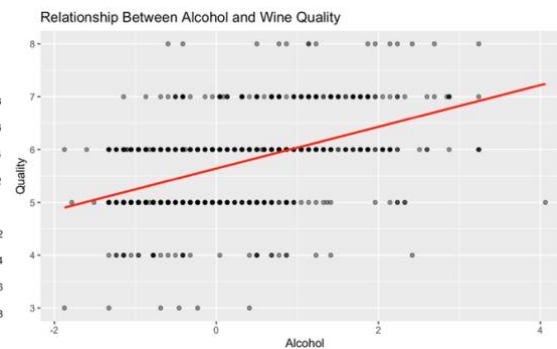


Figure 1.1

1.4 Methodology

1.4.1 Principal Component Analysis (PCA)

Principal Component Analysis (PCA) was conducted on the standardised numeric variables to reduce dimensionality without compromising the main structure of the data. The scree plot (Fig 1.2)* shows a sharp decline in variance explained after the first two principal components, followed by a gradual decrease, indicating an elbow around PC2–PC3. This suggests that the first two to three components capture most of the meaningful variation in the dataset. This observation is confirmed by a supplementary bar plot (Fig 1.3)*. The PCA biplot (Fig 1.4)* of the first two components reveals that variables such as alcohol, sulphates, and acidity-related measures have the strongest influence, as indicated by their longer loading vectors. The directions of these vectors reflect correlations among chemical properties, while the clustering of observations near the origin suggests that most wines exhibit average chemical profiles. Overall, retaining two principal components achieves effective dimensionality reduction while maintaining interpretability of the underlying chemical relationships.

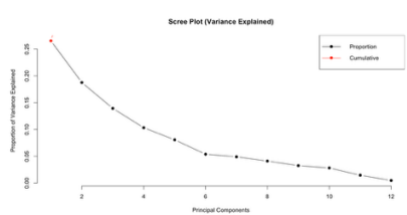


Figure 1.2

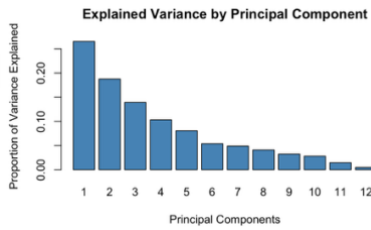


Figure 1.3

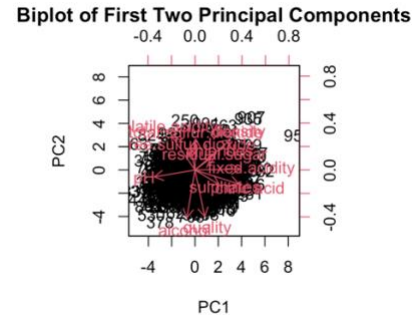


Figure 1.4

1.4.2 K-means Clustering

The first two principal components were used as input for K-means clustering to identify groups of observations with similar characteristics. To select an appropriate number of clusters, the Elbow Method was employed, whereby the total within-cluster sum of squares (WSS) was plotted against different numbers of clusters (k). The Elbow plot (Fig.1.5)* suggests that two to three clusters provide a reasonable balance between model simplicity and explanatory power, and k = 3 is therefore selected for the final clustering.

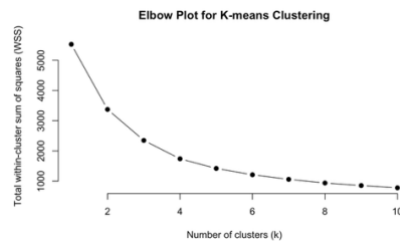


Figure 1.5

1.4.3 Silhouette Analysis

Silhouette analysis is employed to evaluate clustering quality across different combinations of retained principal components and cluster numbers. A score above 0.7 is generally regarded as strong structure, while scores between 0.5 and 0.7 are medium structure and scores between 0.25 and 0.5 are weak structure. Scores below 0.25 suggest no structure (Benaya, Raisa & Sibaroni, Yuliant & Ihsan, Aditya, 2023). Despite the quantitative guidance from silhouette analysis, incorporating our understanding of the data's context remains crucial in deciding the appropriate number of clusters. Analysis shows that retaining 2 PC and forming 2 clusters achieves the highest silhouette score of 0.4 (Fig 1.6)*.

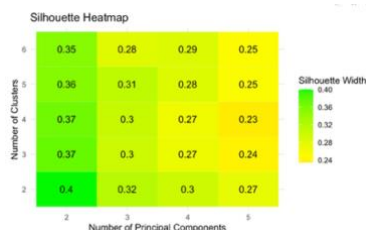


Figure 1.6

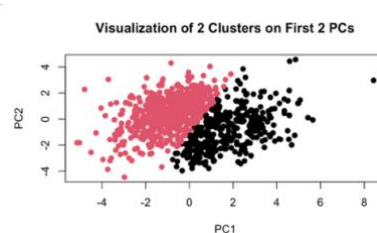


Figure 1.7

1.4.4 Outliers Testing

Outliers are identified using Z-scores to assess their influence on clustering results. K-means clustering is repeated after outlier removal and silhouette scores are compared. As only marginal improvements are observed, outliers are retained since they represent meaningful extreme observations rather than noise.

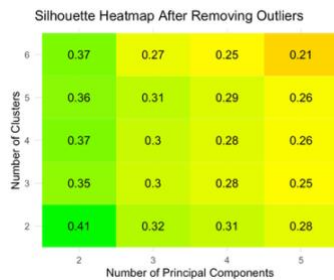


Figure 1.8

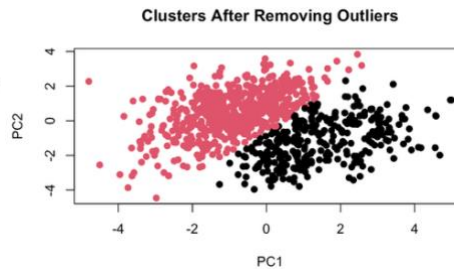


Figure 1.9

1.4.5 Hierarchical Clustering Analysis (HCA)

Hierarchical Cluster Analysis (HCA) is a method that groups objects based on their similarities, creating a dendrogram that illustrates the hierarchy of clustering. It starts by treating each object as a cluster and progressively merges them using criteria like nearest or furthest distance, or minimum variance. This technique enables analysts to identify the number of clusters by applying cuts to the dendrogram at varying levels, providing a flexible and insightful way to explore data relationships. I conducted HCA to evaluate whether Euclidean(Score:0.364^{*}) or Manhattan(Score: 0.286^{*}) distances better captured the structure of our dataset. Euclidean distance is more appropriate for the standardized, continuous PCA scores used in this analysis. I selected Euclidean distance with Average linkage, despite single linkage and average linkage both producing the highest silhouette score (0.548). Average linkage is chosen because it has the highest cophenetic correlation while producing a dendrogram with balanced, interpretable clusters and no chaining effect. This aligns well with earlier K-means results. Although the mean silhouette score of 0.598^{*} indicates that hierarchical clustering does not yield the strongest cluster separation, it remains a valuable exploratory tool for interpreting cluster structure. This is reflected in Figure 1.11, which illustrates meaningful differences in key wine characteristics across the identified clusters.

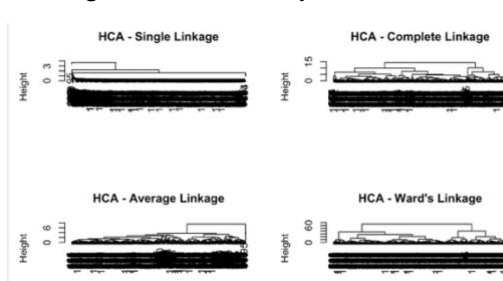


Figure 1.10

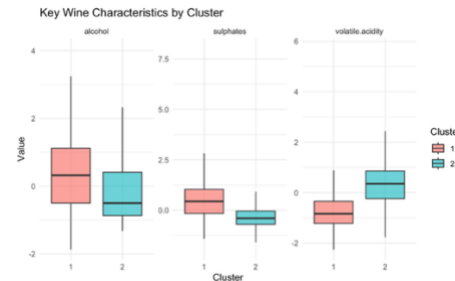


Figure 1.11

1.5 Findings

RQ1: What is the most suitable number of clusters when grouping wines according to their physicochemical properties?

Figures 1.5 & 1.8 show that grouping the wines into two clusters is optimal for our analysis.

RQ2: What insights can be derived from the clusters?

The clustering results reveal meaningful structure within the wine dataset, indicating that wines can be grouped based on shared chemical characteristics without using any prior labels. One cluster is characterized by higher alcohol content and lower volatile acidity, attributes that are positively associated with wine quality. In contrast, the other cluster comprises wines with lower alcohol levels and higher volatile acidity, which are generally associated with decreased quality ratings. This separation suggests that the clustering method effectively groups wines based on meaningful chemical characteristics that align with known quality-related factors. Overall, the results indicate that wine quality is shaped by a combination of physicochemical properties as opposed to any single variable, and that clustering can uncover latent structure in the data that reflects these underlying relationships.

RQ3: How do the identified clusters relate to observed wine quality levels, and do they reveal meaningful patterns not captured by individual variables alone?

The identified clusters exhibit clear differences in observed wine quality levels, with one cluster generally associated with higher quality ratings and the other with lower ratings. These distinctions align with combined physicochemical profiles, such as higher alcohol content and lower volatile acidity in the higher-quality cluster. Importantly, the clustering results reveal multivariate patterns that are not fully captured by individual variables in isolation. Overall, the clustering approach provides a more holistic understanding of wine quality by uncovering latent structures in the data, supporting its value as an exploratory and interpretive analytical tool.

2. Regression-Supervised Learning

2.1 Problem Statement

Building on insights from clustering analysis, this study aims to construct a predictive model that estimates house prices based on key property attributes. The aim is to identify the most influential attribute affecting housing prices, enabling a deeper understanding of the underlying factors driving price variations. This model will not only support accurate price prediction but also provide valuable insights for buyers, sellers, and real estate stakeholders in making informed decisions.

2.2 Research Questions

- **RQ1:** Which property attribute most significantly influence house prices, and which can be considered negligible?
- **RQ2:** Which regression model provides the best fit for the housing dataset, and what does this reveal about the underlying structure of housing price determinants?
- **RQ3:** What are the implications of the regression findings for subsequent studies?

2.3 Dataset Description and Exploratory Data Analysis (EDA)

Key Features Explanations

CRIM: Per capita crime rate within each town.	ZN: Proportion of residential land designated for lots exceeding 25,000 sq.ft.	INDUS: Proportion of land occupied by non-retail business areas within each town.	CHAS: Charles River dummy variable (1=Yes, 0=No).	NOX: Concentration of nitric oxides measured in parts per 10 million.
RM: Mean number of rooms.	AGE: Proportion of owner-occupied units constructed before 1940.	DIS: Weighted distances to five employment centres in Boston.	RAD: Measure of accessibility to radial highways.	TAX: Property-tax rate per \$10,000 of assessed value.
PTRATIO: Student-to-teacher ratio within each town.	B: Proportion of Black residents within each town. (Removed due to ethical concern and bias)	LSTAT: Percentage of population classified as lower SES.	MEDV: Median value of owner-occupied homes, expressed in \$1000's.	

The Wine Quality dataset comprises of 509 rows and 13 columns after data cleaning*. The variable ****B****, representing the proportion of Black residents, was excluded from the cleaned dataset due to ethical concerns and its potential to introduce bias, given that it does not reflect an actionable housing feature. A correlation matrix (Fig 2.0)* was used to identify the strongest linear relationships among variables. MEDV shows a strong positive correlation with RM and a highly negative correlation with LSTAT, with additional moderate negative relationships observed for NOX and PTRATIO. Targeted scatterplots (Fig 2.1)* of these key predictors reinforce these patterns: RM displays a clear positive association with house prices, while LSTAT shows a strong negative trend. NOX and PTRATIO exhibit weaker but still negative relationships with MEDV, suggesting that pollution levels and student-teacher ratios may influence housing values to a lesser extent. Overall, RM and LSTAT emerge as the most influential predictors of MEDV, guiding subsequent modelling decisions.

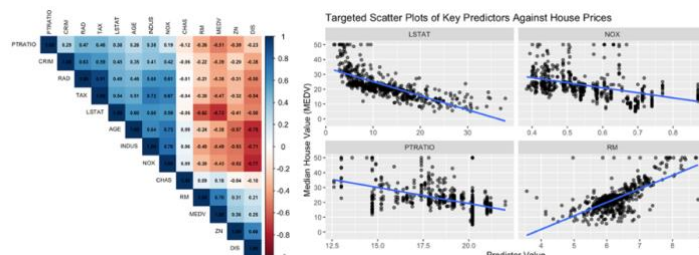


Figure 2.0

Figure2.1

2.4 Methodology

2.4.1 Stepwise Regression

Forward and backward stepwise selection derived from the Akaike Information Criterion (AIC) was applied to balance model fit and complexity, with lower AIC values indicating more preferable models. The stepwise selected model demonstrates strong explanatory power (Adjusted $R^2 = 0.728$), indicating that the selected predictors collectively explain a substantial proportion of the variation in median house value. Regression coefficients quantify the influence of every attribute on the response variable, while standard errors measure estimation precision. Statistical significance is indicated using significance codes, where *** denotes highly significant predictors ($p < 0.001$). Model adequacy is further supported by a low residual standard error (4.789) and a high F-statistic (137 on 10 and 498 DF) with an associated p-value ($< 2.2e-16$) close to zero, confirming overall model significance. Variance Inflation Factor (VIF) diagnostics indicate low to moderate multicollinearity, with most predictors exhibiting VIF values below 5. A Variance inflation factor (VIF) below 5 is not of concern for multicollinearity. Although RAD and TAX show higher VIF values, they remain below the established level of concern ($VIF > 10$), implying that multicollinearity does not materially compromise coefficient stability or inference (Ringim et al., 2024).

```
Call:
lm(formula = MEDV ~ CRIM + ZN + CHAS + NOX + RM + DIS + RAD +
    TAX + PTRATIO + LSTAT, data = dataset_2_fino1)

Residuals:
    Min       1Q   Median       3Q      Max
-15.1367  -2.7877  -0.5896   1.9426  26.4076

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  41.536298   4.891681   8.491 2.37e-16 ***
CRIM         -0.130821   0.032429  -4.034 6.34e-05 ***
ZN           0.048318   0.013680   3.532 0.000451 ***
CHAS         2.843071   0.862674   3.296 0.001052 **
NOX         -18.619464   3.551243  -5.243 2.34e-07 ***
RM           3.675894   0.408375   9.001 < 2e-16 ***
DIS          -1.538116   0.187402  -8.208 1.95e-15 ***
RAD           0.289195   0.063887   4.527 7.50e-06 ***
TAX          -0.012425   0.003405  -3.649 0.000291 ***
PTRATIO      -0.925169   0.130380  -7.096 4.44e-12 ***
LSTAT       -0.537078   0.047210  -11.376 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.789 on 498 degrees of freedom
Multiple R-squared:  0.7334, Adjusted R-squared:  0.728
F-statistic: 137 on 10 and 498 DF, p-value: < 2.2e-16
```

Figure 2.2

Figure 2.3

2.4.2 Regression

Linear Regression captures the relationship between housing attributes and MEDV as a linear function of predictors, providing a simple and interpretable baseline for analysis. **Polynomial Regression** extends this framework by introducing nonlinear terms, allowing the model to show the curvature and more complex patterns present in the housing market. **Ridge Regression** introduces L2 regularization to constrain large coefficients, thereby reducing model variance and enhancing stability when predictors are highly correlated. **LASSO Regression** applies L1 regularization, enabling both coefficient shrinkage and automatic variable selection, resulting in a more parsimonious and interpretable model. **Random Forest** differs fundamentally from these parametric approaches by employing a set of decision trees to model nonlinear relationships and interaction effects. By aggregating predictions across multiple trees, it produces more accurate and robust estimates while reducing overfitting. In addition, Random Forest provides valuable insights into predictor importance. As shown in Figure 2.5*, variables such as RM and LSTAT play a dominant role in predicting housing prices, underscoring the importance of dwelling size and socio-economic conditions. In terms of predictive performance, Random Forest achieves the lowest RMSE among all models evaluated (Fig. 2.4)*, indicating a superior model fit and superior generalization capability relative to the other four models.

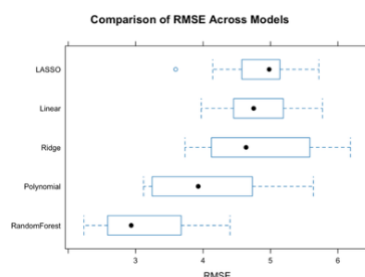


Figure 2.4

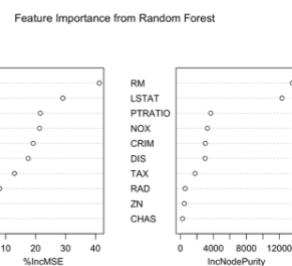


Figure 2.5

2.4.3 Ensemble and Gradient Boosting

A stacking ensemble was carried out using Linear Regression, Random Forest, and Elastic Net as base models. The stacked model, trained using 10-fold cross-validation, resulting in an RMSE of 3.13757(Fig 2.6) *, indicating that combining complementary algorithms improves predictive performance compared to using any single model alone. Following this, I ran Gradient Boosting, capitalizing on its strength in handling complex datasets with nonlinear relationships. The technique improved both RMSE (3.032) * and R-squared values, indicating more accurate predictions and better model fit(Fig. 2.7) *.

```
## The following models were ensemble: lm, rf, glmnet
##
## caret::train model:
## Generalized Linear Model
##
## 500 samples
## 3 predictor
##
## No pre-processing
## Resampling: Cross-Validated (10 fold)
## Summary of sample sizes: 450, 459, 450, 459, 457, 458, ...
## Resampling results:
##
## RMSE   Required MAE
## 3.13757 0.8053289 2.165864
##
## Final model:
##
## Call:  NULL
##
## Coefficients:
## (Intercept)      lm      rf      glmnet
## -1.3162      0.3297      1.0159     -0.2876
##
## Degrees of Freedom: 588 Total (i.e. Null); 585 Residual
## Null Deviance: 42840
## Residual Deviance: 5834 AIC: 2621
```

Figure 2.6

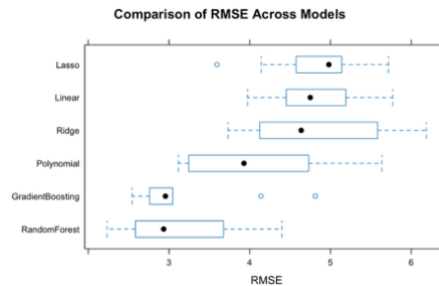


Figure2.7

2.4.4 Tuning

After identifying Random Forest and Gradient Boosting as the strongest-performing models in our analysis, we proceeded to fine-tune their hyperparameters to further improve predictive accuracy. The tuning process involved systematically varying the parameter, which determines how many features are considered at each split and finding that best value. For Random Forest, the optimal configuration was found at $mtry = 6$ *, indicating that using a moderate number of features at each split produced the most stable and accurate predictions (Fig 2.8) *. For Gradient Boosting, the most effective settings were identified as 300 trees, an interaction depth of 3, and a shrinkage value of 0.1, collectively resulting in the lowest RMSE and the most robust model performance (Fig. 2.9, 10) *.

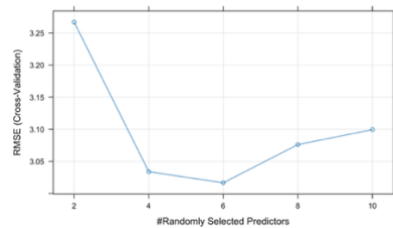


Figure 2.8

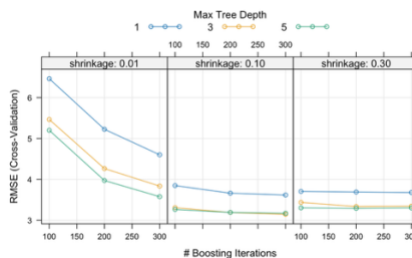


Figure 2.9

##	5	150	3.097049	0.8803118	2.158939
##	5	200	3.032025	0.8857926	2.129541
##	5	250	3.042092	0.8850997	2.133600

Figure 2.10

2.5 Findings

RQ1: Which housing-related variables most significantly influence house prices, and which can be considered negligible?

Stepwise regression identified a subset of predictors that significantly influence MEDV (median house value), indicating that housing prices are driven by a combination of structural, socioeconomic, environmental, and accessibility factors. Variables such as RM (average number of rooms) exhibit a strong positive effect on MEDV, while LSTAT (proportion of lower-status population) shows a pronounced negative effect, confirming these as the most influential predictors. Predictors excluded during the stepwise selection process (e.g. INDUS and AGE) can be considered negligible within this modelling framework, as they did not provide additional explanatory power once more informative variables were included.

Figure 2.5 further supports these findings using Random Forest feature importance measures. The %IncMSE metric represents the percentage increase in prediction error resulting from randomly permuting a given variable, with higher values indicating greater importance to model accuracy. IncNodePurity reflects the cumulative decrease in node impurity associated with a variable across all trees, capturing its overall contribution to the model's decision-making process. Both metrics consistently identify RM and LSTAT as the most important predictors of MEDV. In contrast, ZN, CHAS, and RAD display low importance across both measures and can therefore be regarded as relatively negligible in the Random Forest model.

RQ2: Which regression model provides the best fit for the housing dataset, and what does this reveal about the underlying structure of housing price determinants?

The Ensemble model achieves the best predictive performance, with the lowest RMSE (1.296) and highest R^2 (0.98) (Fig. 2.11)*, indicating it is most effective at modelling median house values. The strong performance of Random Forest and Gradient Boosting, compared to linear and regularised models, suggests that housing prices are driven by nonlinear relationships and complex interactions among predictors rather than simple additive effects (Fig 2.12)*. This implies that key determinants such as housing characteristics, socioeconomic factors, and location-related variables interact in ways that linear models cannot fully capture. Overall, the results reveal that the underlying structure of housing price determinants is complex, nonlinear, and interaction-driven, favouring ensemble and tree-based approaches.

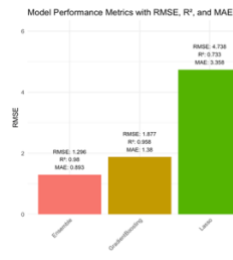


Figure 2.11

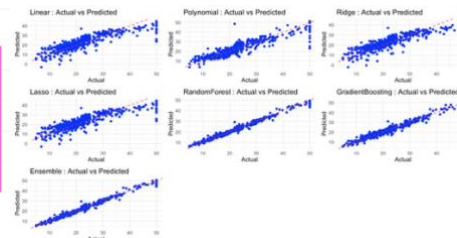


Figure 2.12

RQ3: What are the implications of the regression findings for future research?

Identifying RM and LSTAT as the most influential predictors highlights the importance of structural quality and socioeconomic conditions in determining median house values (MEDV). Improved predictive performance enables a deeper understanding of the key structural, socioeconomic, and locational factors driving housing prices, which can inform urban planning, housing policy, and investment decisions. Research by Babbar, A. (2022) supports the use of machine learning as an effective tool in this endeavour, highlighting its potential to significantly impact the prediction of housing prices.

3. Classification-Supervised Learning

3.1 Problem Statement

Building on evidence that demographic characteristics, lifestyle habits, and respiratory-related symptoms contribute significantly to the development of lung cancer, this project aims to develop machine learning classification models to accurately predict lung cancer diagnosis using patient-level clinical and behavioural features. Early and reliable detection of lung cancer is critical, as the disease is often diagnosed at advanced stages, leading to poor survival outcomes. The proposed models aim to achieve high predictive performance, with particular emphasis on sensitivity to ensure that lung cancer cases are correctly identified, while maintaining sufficient specificity to reduce false positive diagnoses. Achieving this balance supports improved clinical screening, risk stratification, and early intervention, thereby contributing to more effective disease management and better patient outcomes.

3.2 Research Question

- **RQ1:** Which patient characteristics and clinical symptoms are the most important predictors of lung cancer diagnosis?
- **RQ2:** Which machine learning classification model provides the most accurate prediction of lung cancer status?
- **RQ3:** How does class imbalance within the lung cancer dataset affect model performance and predictive accuracy?

3.3 Dataset Description and Exploratory Data Analysis

Gender: Male(M), Female(F)	Age: Age of the patient in years, representing a key demographic risk factor.	Smoking: whether the patient smokes, a primary risk factor for lung cancer.	Yellow Fingers: Presence of yellow staining on fingers, often associated with long-term smoking.
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Anxiety: Indicates whether the patient experiences anxiety	Peer Pressure: Reflects social influence that may encourage smoking behaviour.	Chronic Disease: Indicates the presence of long-term illnesses that may increase vulnerability to lung cancer.	Fatigue: Represents persistent tiredness, a common symptom reported in chronic illness.
Allergy: Indicates whether the patient has allergies, which may affect respiratory health.	Wheezing: Presence of wheezing sounds during breathing, often linked to respiratory conditions.	Alcohol: Indicates alcohol consumption, a lifestyle factor potentially associated with cancer risk.	Coughing: Presence of persistent coughing, a common symptom in lung-related diseases.
Shortness of Breath: Indicates breathing difficulty, often associated with lung impairment.	Swallowing Difficulty: Reflects discomfort or difficulty swallowing, which may indicate advanced disease symptoms.	Chest Pain: Presence of chest pain, a symptom frequently associated with lung and respiratory conditions.	Lung Cancer: Binary target variable indicating the presence or absence of a lung cancer diagnosis.

The Lung Cancer dataset (Ahmad Bhat, 2023) comprises of 509 rows and 13 columns after data cleaning^{*}. It includes no synthetic records and is devoid of missing data. As is typical with medical datasets, it displayed significant class imbalance (Fig. 3.0)^{*}. To mitigate this issue and enhance classification performance, the Synthetic Minority Over-sampling Technique (SMOTE) was utilised to create additional samples belonging to the minority class, which has proven to be successful at enhancing performance on unbalance dataset(GeeksforGeeks, 2025). Exploratory analysis and variance inflation factor (VIF) assessment indicated no severe multicollinearity among predictors, ensuring model stability and interpretability.

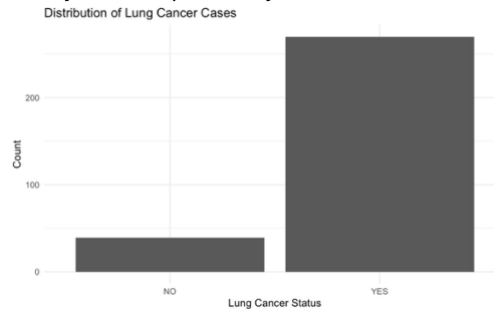


Figure 3.0

3.4 Methodology

3.4.1 Classification

Model Introduction: **Logistic Regression** is a commonly applied statistical learning approach to compute the probability of a binary outcome from a set of explanatory variables. **Random Forest** is an ensemble learning technique that combines multiple decision trees to improve prediction accuracy, offering strong performance, robustness to overfitting, and the ability to handle complex feature interactions. **XGBoost (eXtreme Gradient Boosting)** is a powerful boosting algorithm that sequentially constructs decision trees, with each iteration emphasising the correction of errors from prior models, and is recognised for its computational efficiency and high predictive accuracy. **K-Nearest Neighbours (KNN)** is a non-parametric classification technique that assigns class labels based on similarity to neighbouring observations, making it effective for problems with irregular decision boundaries. **Support Vector Machines (SVM)** aims to determine the optimal separating hyperplane between classes and are particularly well in high-dimensional spaces, though they can be computationally intensive.

3.4.2 Performance Metrics

3.4.2.1 Accuracy

Accuracy represents the proportion of correctly classified cases, encompassing both true positives (correctly predicted patients) and true negatives (correctly predicted non-patients), with respect to the total number of observations. It serves as a clear and easily interpretable measure of a model's overall classification performance. From Figure 3.1^{*}, XGBoost demonstrate commendable performance in the Accuracy metrics component.

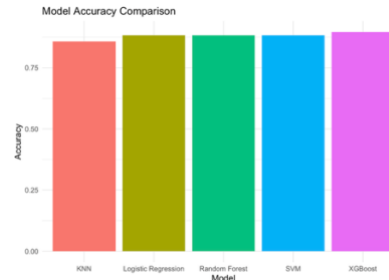


Figure 3.1

3.4.2.2 F1-Score

The F1 Score captures the harmonic mean of precision and recall, ensuring a balance between them, especially useful when classes are unevenly distributed or when the consequences of false positives differs from false negatives. This metric is notably valuable in medical diagnostics, where failing to detect a condition (low recall) or falsely alarming a condition (low precision) can have severe implications. From Figure 3.2*, XGBoost show strong performance in F1 Score. The confusion matrix provides the basis for computing both accuracy and the F1 score (Fig. 3.3)*. Figure 3.4 explains how to read the confusion matrix (Suffled, M, 2019).

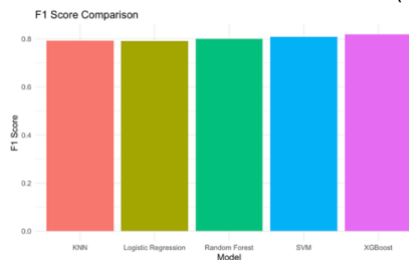


Figure 3.2

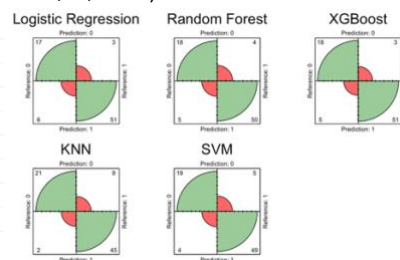


Figure 3.3

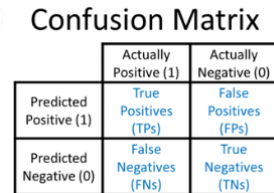


Figure 3.4

3.4.2.3 Area Under the Receiver Operating Characteristic Curve (ROC AUC)

The ROC AUC score is a way to assess how well a model can distinguish between categories. The ROC is a curve that plots the relationship between sensitivity and specificity at varying cut-off levels. The AUC shows the model's ability to correctly classify the groups. Essentially, by analysing the entire area below the ROC curve, the ROC AUC score gives an overall performance rating for the model across all potential decision thresholds, making it a comprehensive measure of effectiveness. XGBoost (Fig.3.5 & 6)* performs exceptionally well, suggesting it effectively balances false positives and false negatives, thereby rendering it a suitable choice for varied threshold levels.

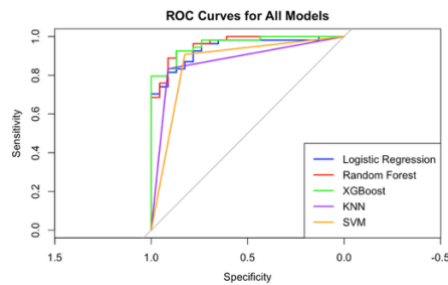


Figure 3.5

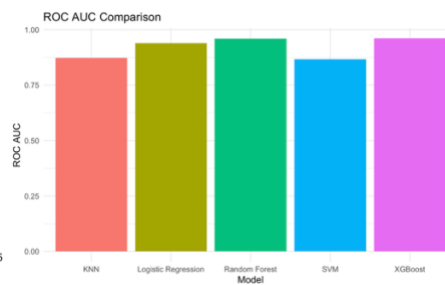


Figure 3.6

3.4.3 Tuning

We find out best tuning parameter to find the optimal configuration of setting that results in the best performance of a given model. After finding out the best parameters* for each model, for example Random Forest adjusting “mtry” parameter to 5, maintaining model stability (McNemar’s Test p-value: 0.7237)*, indicating no statistically significant difference between the compared model predictions. This indicates tuning maintained the model’s stability and did not negatively affect its performance. Overall, post-tuning, the models demonstrated improved or maintained high performance across various metric (Fig. 3.7, 8, & 9).



Figure 3.7



Figure 3.8

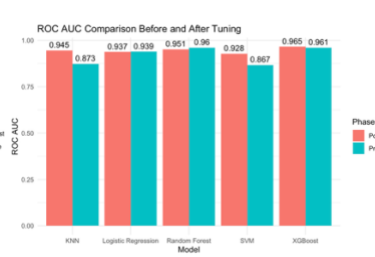


Figure 3.9

3.5 Findings

RQ1: Which patient characteristics and clinical symptoms are the most important predictors of lung cancer diagnosis?

The most significant predictor for lung cancer in the dataset is alcohol consumption, highlighting the strong influence of lifestyle-related factors on lung cancer diagnosis. This is followed closely by allergy history and difficulty swallowing, which reflect underlying respiratory and immunological conditions associated with the disease. Additional important predictors include coughing, wheezing, and chronic disease, emphasizing the relevance of persistent respiratory symptoms and comorbidities in identifying lung cancer cases. Demographic and behavioural factors such as age and peer pressure contribute moderately to predictive performance, while variables including gender, smoking, and shortness of breath show comparatively lower importance within this dataset. Future research could focus on further investigating the interaction between lifestyle behaviours and clinical symptoms, as well as incorporating longitudinal patient data to enhance early detection and risk stratification strategies (Fig. 3.10)*.

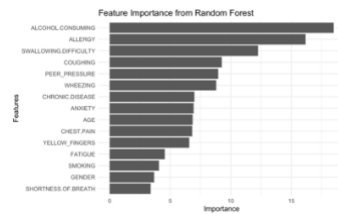


Figure 3.10

RQ2: Which machine learning classification model provides the most accurate prediction of lung cancer status?

Before tuning, XGBoost demonstrated the most reliable and balanced classification performance for lung cancer prediction, demonstrating strong accuracy, F1 score, and high discriminative ability. After tuning, Random Forest is the most reliable model, delivering consistently strong performance across all metrics with minimal sensitivity to tuning. XGBoost provides a competitive alternative with stable results and efficient training. Overall, XGBoost is selected as the preferred model, as it demonstrates the most consistent and reliable performance both before and after hyperparameter tuning (Fig 3.7, 8, 9, 11)*.

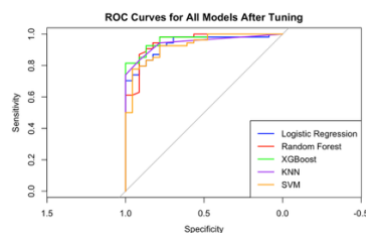


Figure 3.11

RQ3: How does the imbalance within the lung cancer dataset impact the accuracy of our prediction model?

The use of SMOTE to mitigate class imbalance in the lung cancer dataset may introduce challenges such as potential overfitting, as synthetic samples replicate patterns from the minority class. This replication can oversimplify the underlying data structure, which is particularly concerning in lung cancer prediction, where clinical symptoms and risk factors can vary widely in onset, severity, and presentation. While SMOTE improves class balance and supports effective model training, it may limit the model's ability to capture small but clinically meaningful variations among lung cancer cases. Moreover, recent research suggests that artificially correcting class imbalance may negatively affect model calibration and real-world validity by altering the natural data distribution (Molnar, 2023). Future research should therefore explore more advanced modelling approaches that are robust to class imbalance without reliance on resampling techniques, enabling improved generalisation and more accurate representation of complex clinical patterns (Seifollah Gholampour, 2024).

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